

CN4119 Final Year Design Project

Cost Estimation and Economic Analysis - Introduction

Dr Suraj Vasudevan

E4 06-03

Email: chesura@nus.edu.sg

Recommended Readings

Turton et al.'s book (Reference 5) Chapters 7-10

Contents

- Introduction
- Cost estimation
 - Estimation of capital cost
 - Estimation of manufacturing cost
- Economic analysis
 - Engineering economic analysis
 - Profitability analysis

Learning Outcomes

You should be able to:

1. Describe

- Estimation of capital and operating costs

2. Estimate

- Capital and operating costs of equipment and plants

3. Describe

- Time value of money, cash flow diagram, depreciation and profitability criteria

4. Perform

- Economic analysis of equipment and plants

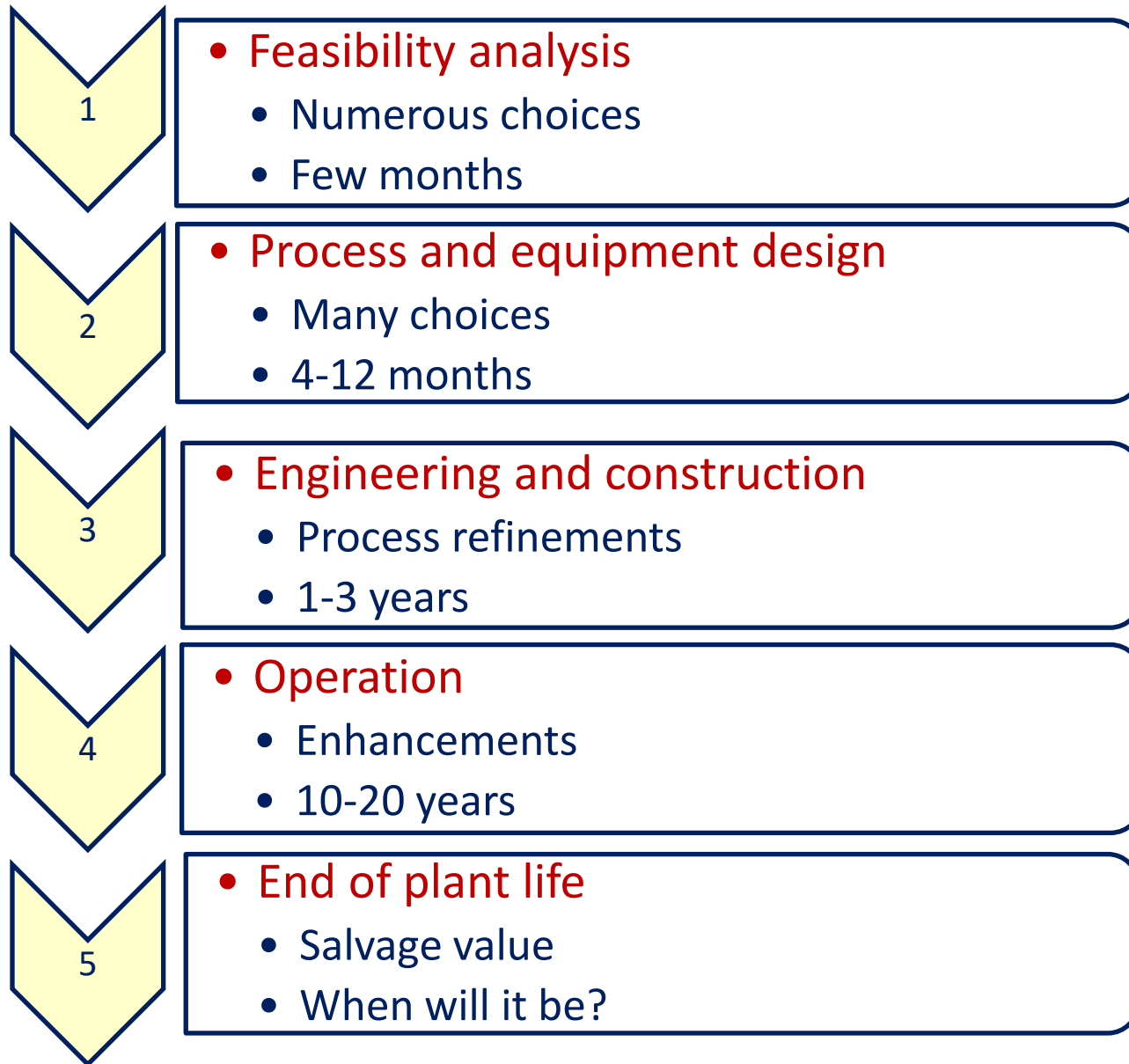
Introduction

❑ Key issues for consideration

- Investing in a new plant and/or modifying an existing plant?
- Greenfield or developed site?
- Plant operation for many years?



Life Cycle of a Process Plant



Cost Estimation and Economic Analysis

- ❑ Why does a company invest?
- ❑ Is some risk involved in the investment?
- ❑ When is cost estimation and economic analysis required?
- ✓ What is cost estimation?
- ✓ What is economic analysis?
- ✓ How do you estimate the cost and evaluate profitability?

Cost estimation and economic analysis
are required for ***Sustainability
Assessment***

